Lab 5 ticket

Long-Run Mix and Eigen — Friday, November 20

Lab 5 — Long-Run Mix and Eigen · Fri Nov 20 · ~25–30 min · Laptop · Handwrite · Initials

Name: _____ **Language (circle):** Octave / Python

New syntax

I needed to…	Command / pattern
Loop $x \leftarrow Ax$	
Eigenpairs of A	
Scale / normalize v	

$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$. Quiz 10 = Week 11 (**not** this lab).

Part A — Discover

C5.0. $x_0 = (1000, 0)$. $x_1 =$ _____ $x_5 \approx$ _____ $x_{20} \approx$ _____
Expected: $x_1 = (700, 300)$; settle near $(400, 600)$.

C5.1. From $x_0 = (0, 1000)$, after 20 steps: $\approx$ _____
Surprising? _____

Part B — Explain

C5.2. Eigenvalues: _____ $|\lambda| = 1$? _____ $|\lambda| < 1$? _____

C5.3. Scaled $\lambda = 1$ vector (sum 1000): _____ vs x_{20}: _____
Why settle there? _____

C5.4. (If time.) Capstone link: _____

Stuck on: _____

Check-in initials: _____